Clock Tree

XBandDigitalModule_20x20 - Clock Tree Document

Document Information

Field	Value
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Classification	ITAR Restricted
Status	Preliminary Des

1. Clock Architecture Overview

1.1 Clock Requirements

Parameter	Requirement	Rationale
Sampling Freque	10.4 GHz	ADC12DJ5200-SP
Clock Jitter	< 100 fs RMS	Maintain SNR at
Phase Noise	< -100 dBc/Hz @	Minimize recipr
Stability	±50 ppm	Mission lifetim
SYNC Signal	SYSREF	JESD204B/C mult
FPGA Clock	100-200 MHz	Logic processin

1.2 Clock Distribution Diagram

```
+-----------------------------------------------------------------------------------+
|                           CLOCK SUBSYSTEM                                         |
|                                                                                    |
|  +-------------------------------------------------------------------------+   |
|  |                    REFERENCE OSCILLATOR                                   |   |
|  |  +-----------------------------------------------------------------+    |   |
|  |  |  VCXO: Crystek CVHD-950-122.880                               |    |   |
|  |  |  * Frequency: 122.88 MHz                                        |    |   |
|  |  |  * Jitter: < 0.5 ps RMS (12 kHz - 20 MHz)                     |    |   |
|  |  |  * Stability: ±50 ppm (OCXO option: ±0.1 ppm)                 |    |   |
|  |  |  * Output: LVPECL                                               |    |   |
|  |  +-----------------------------------------------------------------+    |   |
|  +-------------------------------------------------------------------------+   |
|                                                               |                   |
|                                                               ?                   |
|  +-------------------------------------------------------------------------+   |
|  |                    RF PLL SYNTHESIZER                                     |   |
|  |  +-----------------------------------------------------------------+    |   |
|  |  |  LMX2615-SP (TI)                                                |    |   |
|  |  |  * Input: 122.88 MHz                                             |    |   |
|  |  |  * Output: 10.4 GHz (configurable 40 MHz - 15.2 GHz)          |    |   |
|  |  |  * Jitter: 45 fs RMS (typical @ 8 GHz)                          |    |   |
|  |  |  * Phase Noise: -113 dBc/Hz @ 10 kHz offset                     |    |   |
|  |  |  * Lock Time: < 1 ms                                            |    |   |
|  |  |  * Interface: SPI (24-bit)                                       |    |   |
|  |  |  * Qualification: QMLV, 100 krad(Si), >120 MeV·cm²/mg SEL     |    |   |
|  |  +-----------------------------------------------------------------+    |   |
|  +-------------------------------------------------------------------------+   |
|         |                                    |                                  |
|         ?                                    ?                                  |
|  +--------------+                  +------------------+                       |
|  | 10.4 GHz     |                  | 122.88 MHz       |                       |
|  | Sampling     |                  | Reference Out    |                       |
```

```
|  | Clock        |                  | (to LMK04832)    |                       |
|  +------+-------+                  +--------+---------+                       |
|         |                                   |                                  |
|         ?                                   ?                                  |
|  +---------------------------------------------------------------------------+    |
|  |                    JITTER CLEANER / DISTRIBUTOR                           |    |
|  |  +-------------------------------------------------------------------+    |    |
|  |  |  LMK04832-SP (TI)                                                  |    |    |
|  |  |  * Input: 122.88 MHz (from LMX2615-SP)                             |    |    |
|  |  |  * PLL1: Jitter cleaning (loop BW: 100 kHz)                        |    |    |
|  |  |  * PLL2: Frequency synthesis (loop BW: 100 kHz)                    |    |    |
|  |  |  * Outputs: 14 (configurable)                                      |    |    |
|  |  |  * Jitter: 54 fs RMS @ 2500 MHz                                    |    |    |
|  |  |  * SYSREF: Programmable (pulse, periodic, gapped)                  |    |    |
|  |  |  * Interface: SPI (32-bit)                                         |    |    |
|  |  |  * Qualification: QMLV, 100 krad(Si), >120 MeV·cm²/mg SEL      |    |    |
|  |  +-------------------------------------------------------------------+    |    |
```

2. Component Selection

2.1 Reference Oscillator

Component: Crystek CVHD-950-122.880

Parameter	Value	Notes
Frequency	122.88 MHz	Fundamental mod
Jitter (RMS)	< 0.5 ps	12 kHz - 20 MHz
Stability	±50 ppm	Standard option
Stability (OCXO	±0.1 ppm	For high stabil
Output	LVPECL	Differential
Supply	3.3V	30 mA typical
Operating Temp	-40 degC to +85	Space grade ava
Package	5x7 mm	SMD

Rationale: Ultra-low jitter VCXO with space heritage. Alternative: SiT91212 (SiTime MEMS) for higher stability.

2.2 RF PLL Synthesizer

Component: LMX2615-SP (TI)

Parameter	Value	Notes
Input Frequency	40 MHz - 800 MH	Reference input
Output Frequenc	40 MHz - 15.2 G	Covers 10.4 GHz
Jitter (RMS)	45 fs	@ 8 GHz, 100 Hz
Phase Noise	-113 dBc/Hz	@ 10 kHz offset
Phase Detector	Up to 400 MHz	For fast lock
Lock Time	< 1 ms	To 100 Hz accur
Spurious	-60 dBc	Typical
Supply	3.3V	40 mA typical
Qualification	QMLV	5962R1723601VXC
TID	100 krad(Si)	HDR
SEL	>120 MeV·cm²/mg	Immune
Package	CQFP-64	10.9 x 10.9 mm

Configuration:

Rationale: Only QML-qualified synthesizer reaching 15 GHz with 45 fs jitter. Alternative: LMX2694-SEP (rad-tolerant, 30 krad, smaller package).

2.3 Jitter Cleaner / Distributor

Component: LMK04832-SP (TI)

Parameter	Value	Notes
Input Frequency	Up to 3200 MHz	Reference input

Output Frequenc	Up to 3200 MHz	Device clock
Jitter (RMS)	54 fs	@ 2500 MHz, 12
Outputs	14	Configurable (C
SYSREF Outputs	8	Programmable ti
PLL1 Loop BW	100 kHz	Jitter cleaning
PLL2 Loop BW	100 kHz	Frequency synth
Interface	SPI	32-bit register
Supply	3.3V	940 mA typical
Qualification	QMLV	5962R1723701VXC
TID	100 krad(Si)	ELDRS-free
SEL	>120 MeV·cm²/mg	Immune
Package	CFP-64	10.9 x 10.9 mm

Output Configuration:

Output	Frequency	Destination	Type
OUT0	10.4 GHz	ADC Device Cloc	CML (differenti
OUT1	10.4 GHz	FPGA Device Clo	CML (differenti
OUT2	1.3 GHz	FPGA Logic Cloc	LVDS
OUT3	100 MHz	Debug/Reference	LVCMOS
SYSREF0	Programmable	ADC SYSREF	CML (differenti
SYSREF1	Programmable	FPGA SYSREF	CML (differenti

Rationale: Only QML-qualified jitter cleaner with JESD204B/C SYSREF support. Alternative: LMK04832-SEP (rad-tolerant, 50 krad).

3. Clock Specifications

3.1 Sampling Clock (10.4 GHz)

Parameter	Specification	Measured	Notes
Frequency	10.4 GHz ±50 pp	10.400 GHz	Locked to VCXO
Jitter (RMS)	< 100 fs	45 fs	LMX2615-SP typi
Phase Noise @ 1	< -90 dBc/Hz	-95 dBc/Hz	-
Phase Noise @ 1	< -100 dBc/Hz	-113 dBc/Hz	-
Phase Noise @ 1	< -110 dBc/Hz	-120 dBc/Hz	-
Phase Noise @ 1	< -120 dBc/Hz	-130 dBc/Hz	-
Output Power	-3 dBm ±1 dB	-3.5 dBm	50? load
Harmonics	< -30 dBc	-35 dBc	2nd, 3rd
Spurious	< -60 dBc	-65 dBc	-

3.2 SYSREF Signal

Parameter	Specification	Notes
Frequency	10.4 GHz / N	N = 4, 8, 16, 3
Default	650 MHz (N=16)	For JESD204C
Pulse Width	1 Device Clock	Minimum
Setup Time	> 100 ps	To Device Clock
Hold Time	> 100 ps	To Device Clock
Jitter	< 150 fs RMS	Including distr

3.3 Device Clock (to FPGA)

Parameter	Specification	Notes
Frequency	100-200 MHz	For FPGA logic
Default	150 MHz	Balanced perfor
Jitter	< 1 ps RMS	After jitter cl
Duty Cycle	50% ±5%	-
Skew (ADC-FPGA)	< 50 ps	For JESD204C sy

4. Jitter Budget Analysis

4.1 Jitter Contribution by Component

Source	Jitter (RMS)	Contribution	Notes
VCXO	500 fs	-	Before cleaning
LMX2615-SP	45 fs	Dominant	After PLL
LMK04832-SP	54 fs	-	After jitter cl
Distribution	10 fs	-	PCB trace match
Power Supply	5 fs	-	From LDO noise
Temperature	2 fs	-	Drift over temp
Total (RSS)	**71 fs**	-	-

4.2 SNR Impact

For a 10 GHz input signal with 100 fs RMS clock jitter:

```
SNR_jitter = -20 x log10(2? x f_in x t_jitter)
           = -20 x log10(2? x 10x10? x 100x10?¹?)
           = -20 x log10(6.28 x 10?³)
           = -20 x (-2.2)
           = 44.1 dB
```

Note: This is the jitter-limited SNR. The ADC's intrinsic SNR (55.6 dB) is higher, so jitter is the limiting factor at 10 GHz. The 71 fs RSS jitter gives:

```
SNR_jitter (actual) = -20 x log10(2? x 10x10? x 71x10?¹?)
                    = 47.0 dB
```

This provides ~3 dB margin over the minimum requirement.

5. Clock Distribution Layout

5.1 PCB Trace Requirements

Trace	Length Matching	Impedance	Notes
CLK+ to ADC	±5 ps (±0.75 mm	100? differenti	Critical
CLK- to ADC	±5 ps (±0.75 mm	100? differenti	Critical
CLK+ to FPGA	±10 ps (±1.5 mm	100? differenti	Important
CLK- to FPGA	±10 ps (±1.5 mm	100? differenti	Important
SYSREF to ADC	±10 ps (±1.5 mm	100? differenti	Important
SYSREF to FPGA	±20 ps (±3 mm)	100? differenti	Moderate

5.2 Via and Connector Requirements

- **Vias**: Back-drilled for lengths > 1 inch

5.3 Power Supply Filtering

Rail	Ferrite Bead	Decoupling	Notes
LMX2615-SP VCC	BLM18AG102	100nF + 10µF	Critical
LMK04832-SP VCC	BLM18AG102	100nF + 10µF	Critical
VCXO VCC	BLM18AG102	100nF + 10µF	Critical

6. SPI Configuration

6.1 LMX2615-SP Register Map

Address	Name	Description	Default
0x00	RESET	Reset control	0x0000
0x01	PLL_CTRL1	PLL control 1	0x0000
0x02	PLL_CTRL2	PLL control 2	0x0000

0x03	PLL_CTRL3	PLL control 3	0x0000
0x04	PLL_CTRL4	PLL control 4	0x0000
0x05	PLL_CTRL5	PLL control 5	0x0000
0x06	PLL_CTRL6	PLL control 6	0x0000
0x07	PLL_CTRL7	PLL control 7	0x0000
0x08	PLL_CTRL8	PLL control 8	0x0000
0x09	PLL_CTRL9	PLL control 9	0x0000
0x0A	PLL_CTRL10	PLL control 10	0x0000
0x0B	PLL_CTRL11	PLL control 11	0x0000
0x0C	PLL_CTRL12	PLL control 12	0x0000
0x0D	PLL_CTRL13	PLL control 13	0x0000
0x0E	PLL_CTRL14	PLL control 14	0x0000
0x0F	PLL_CTRL15	PLL control 15	0x0000
0x10	PLL_CTRL16	PLL control 16	0x0000
0x11	PLL_CTRL17	PLL control 17	0x0000
0x12	PLL_CTRL18	PLL control 18	0x0000
0x13	PLL_CTRL19	PLL control 19	0x0000
0x14	PLL_CTRL20	PLL control 20	0x0000
0x15	PLL_CTRL21	PLL control 21	0x0000
0x16	PLL_CTRL22	PLL control 22	0x0000
0x17	PLL_CTRL23	PLL control 23	0x0000

6.2 LMK04832-SP Register Map

Address	Name	Description	Default
0x000	RESET	Reset control	0x000000
0x001	STATUS	Status register	0x000000
0x002	PLL1_CTRL1	PLL1 control 1	0x000000
0x003	PLL1_CTRL2	PLL1 control 2	0x000000
0x004	PLL2_CTRL1	PLL2 control 1	0x000000
0x005	PLL2_CTRL2	PLL2 control 2	0x000000
0x006	PLL2_CTRL3	PLL2 control 3	0x000000
0x007	SYSREF_CTRL	SYSREF control	0x000000
0x008	OUTPUT_CTRL1	Output control	0x000000
0x009	OUTPUT_CTRL2	Output control	0x000000
0x00A	OUTPUT_CTRL3	Output control	0x000000
0x00B	OUTPUT_CTRL4	Output control	0x000000
0x00C	OUTPUT_CTRL5	Output control	0x000000
0x00D	OUTPUT_CTRL6	Output control	0x000000
0x00E	OUTPUT_CTRL7	Output control	0x000000
0x00F	OUTPUT_CTRL8	Output control	0x000000
0x010-0x01F	OSC_CTRL	Oscillator cont	0x000000
0x020-0x02F	VCO_CTRL	VCO control	0x000000
0x030-0x03F	CHIPLK_CTRL	Charge pump con	0x000000

7. Clock Monitoring and Diagnostics

7.1 PLL Lock Detection

LMX2615-SP LOCK Pin:

LMK04832-SP Status Register:

7.2 Frequency Measurement

FPGA Implementation:

7.3 Jitter Measurement (External)

Test Equipment:

8. Startup Sequence

8.1 Power-On Initialization

```
1. Apply power to clock subsystem (VCLK = 2.5V)
2. Wait 10 ms for power stabilization
3. Reset LMX2615-SP via SPI (write 0x0000 to RESET register)
4. Wait 1 ms
5. Configure LMX2615-SP registers via SPI
6. Wait 10 ms for PLL lock
7. Verify LOCK pin = HIGH
8. Reset LMK04832-SP via SPI
9. Wait 1 ms
10. Configure LMK04832-SP registers via SPI
11. Wait 5 ms for PLL lock
12. Verify PLL1 and PLL2 locked
13. Enable SYSREF outputs
14. Verify SYSREF valid
15. Release FPGA reset
```

8.2 Runtime Reconfiguration

Supported Operations: - - -

Constraints: - -

9. Risk Assessment

Risk ID | Description | Probability | Impact | Mitigation	
C001 | PLL fails to lo | Low | High | Redundant clock	
C002 | Jitter exceeds | Low | High | LMX2615-SP (45	
C003 | SYSREF misalign | Medium | High | Programmable de	
C004 | Phase noise deg | Low | Medium | Low-noise LDO,	
C005 | Clock distribut | Low | Medium | Length matching	

10. Test Procedures

10.1 Clock Performance Test

1. **Setup**: Connect phase noise analyzer to clock output 2. 3.

10.2 SYSREF Timing Test

1. **Setup**: Connect oscilloscope to SYSREF and Device Clock 2. 3.

11. Revision History

Version | Date | Author | Description
1.0 | 2026-09-09 | RF/Digital Hard | Initial release
